

# A Fuzzy Decision Support System to Optimize Irrigation Practices in Trentino Region

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# Introduction

## The Need for Smarter Irrigation



- **Sustainable water use is essential** to address climate change, water scarcity, and rising productivity demands
- High value crops require **precise soil moisture control** to protect yield and quality
- **Traditional irrigation** methods are often **inefficient** and **non-adaptive** to real field conditions

Figure: Overhead Sprinkler Irrigation of Vineyards in Trentino, Italy

# Introduction

## Solution & Research Objectives

*“ A **Decision Support System (DSS)** is a system that supports decision-making by transforming data and knowledge into clear, actionable recommendations ”*



### Intelligent Decisions

Provide daily, interpretable irrigation recommendations, integrating multiple sources



### Optimization

Calibrate the system based on expert feedbacks, improving decision quality



### Validation

Evaluate the performance using a consistent methodology

# Study Overview

## Use Case: Rovere della Luna consortium

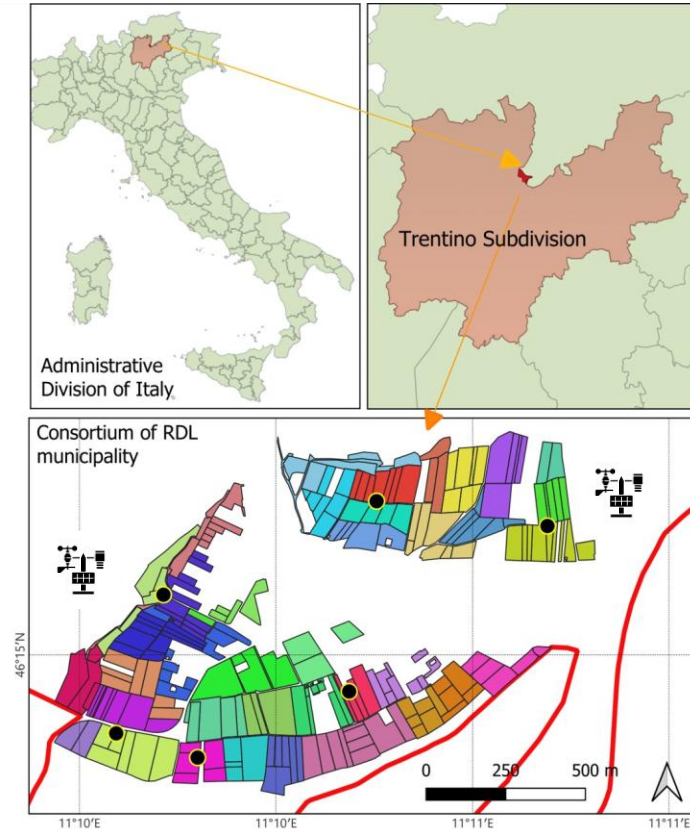
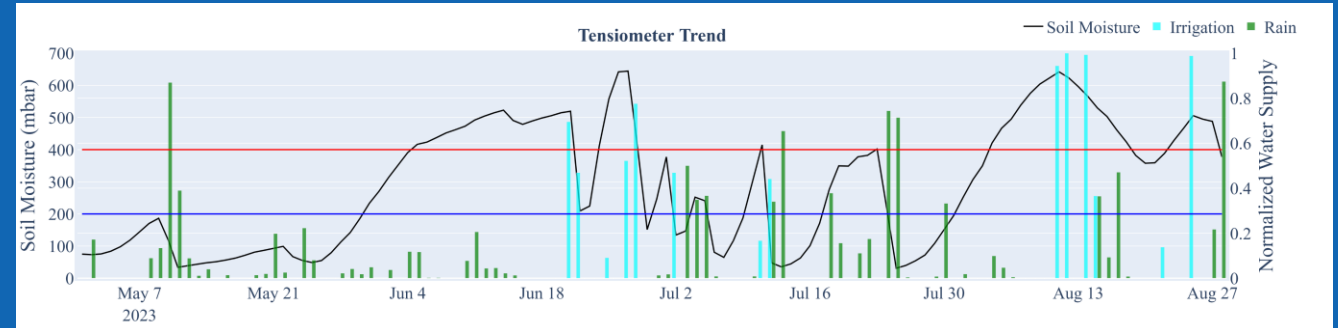


Figure: Map showing the geographical area of the Rovere della Luna consortium, with tensiometers and weather stations



### Consortium Characteristics:

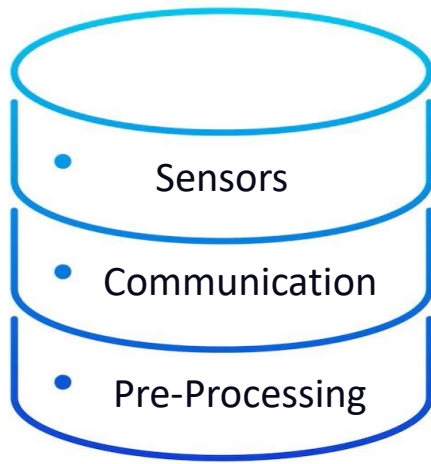
- ❖ Location: Rovere della Luna consortium, Trentino, Italy
- ❖ Period: Summer 2023 (May – September)
- ❖ Crop Type: High-value vineyards
- ❖ Sensor Network:
  - 16 tensiometers placed at 30cm and 60cm depths
  - 2 weather stations (temperature, humidity, rainfall, radiation)

➡ 5 sectors selected for the study

# Study Overview

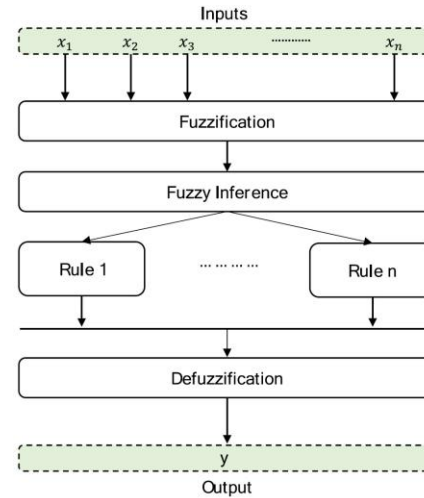
## Decision Workflow

1-3



Edge & Data Ingestion

4



Decision Support System

5



Irrigation Scheduling

# DSS Architecture

## Variables & Fuzzification

### Input Variables:

- 💧 Last Average Tensiometer – previous day (mbar)
- 🤖 Predicted Average Tensiometer – via LSTM (mbar)
- ☁️ Predicted Rain Amount – 3 days (mm)
- 🌡️ Predicted Maximum Temperature – 3 days (°C)

⇒ 3 linguistic terms (Low, Medium, High)

### Output:

- 🌿 Irrigation Recommendation – Decision

⇒ 4 linguistic terms (Number of Irrigation Turns)

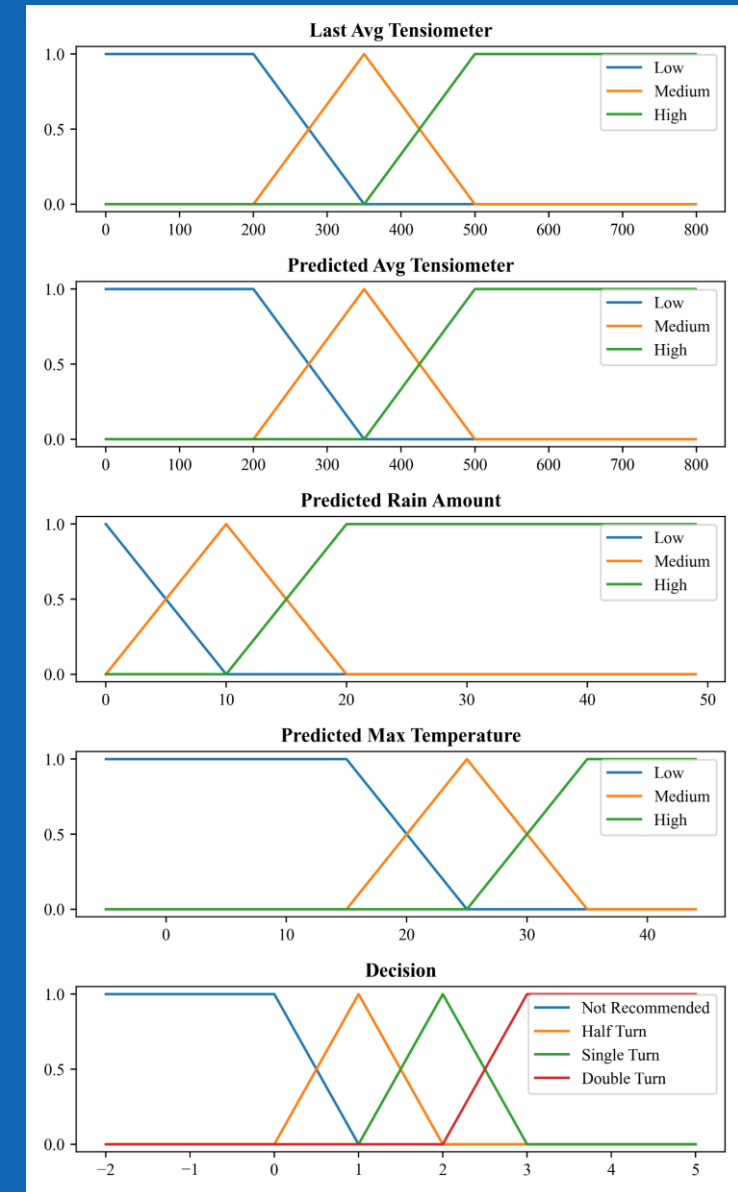


Figure: Fuzzy linguistic variables of the Fuzzy DSS

# DSS Architecture

## Expert Rules and Inference Logic

The Fuzzy DSS makes irrigation decisions using **21 “IF–THEN” rules** based on agronomic expertise  
It uses a **Mamdani**-type inference approach with centroid defuzzification method

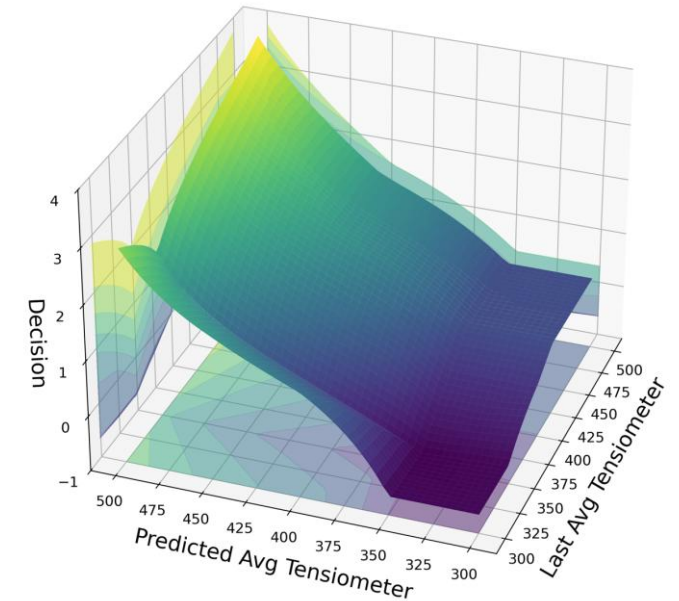
Example Rules:

IF Predicted Tensiometer IS High AND Predicted Rain IS Low → Double Turn

IF Predicted Tensiometer IS Low AND Max Temperature IS Low → No Irrigation

Inference Logic:

📥 Input (Fuzzified) → ⚙️ Rules Evaluation → 🎯 Defuzzification



# DSS Architecture

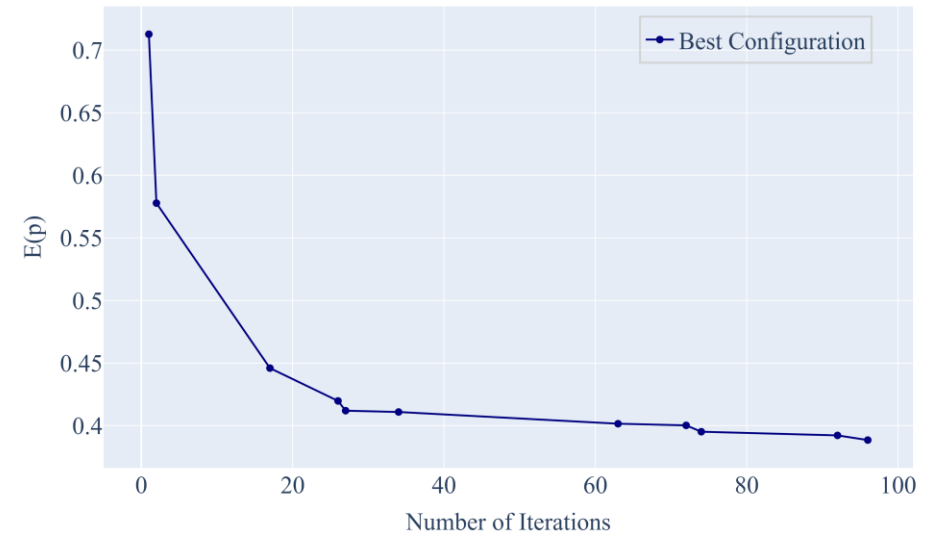
## Bayesian Optimization with Feedback

The system was calibrated using expert feedback, collected via surveys, and **Bayesian optimization** to fine-tune fuzzy membership functions for improving alignment with human recommendations

### Error Minimization Function (MSE)

$$E(p) = \frac{1}{N} \sum_{i=1}^N (y_i(p) - \hat{y}_i)^2$$

- $y_i(p)$ : current DSS output for the  $i$ -th input
- $\hat{y}_i$ : corresponding expert's recommendation
- $p$ : fuzzy membership parameters
- $N$ : number of evaluation scenarios



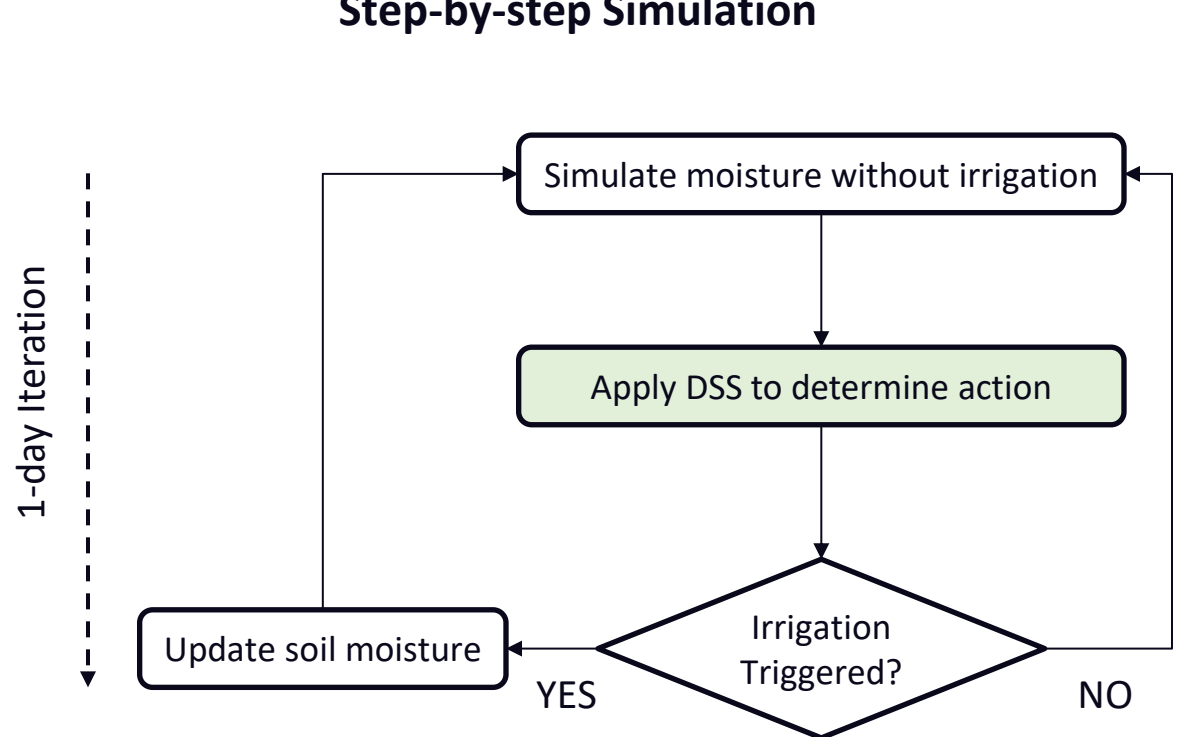
⇒ Error reduced by 45% after 100 iterations

# Validation Framework

## Evaluation Setup and Performance Metrics

The irrigation strategies were tested through **day by day simulation** of soil moisture over the 2023 season across 5 vineyard sectors, using **30 cm tensiometer** data as reference

### Step-by-step Simulation



### Evaluation Metrics

-  Total water used (liters per row)
-  Critical dry days – soil above threshold (400 mBar)
-  Average soil moisture value (mBar)

# Results

## Sector-Level Simulation: Observed vs Recommended

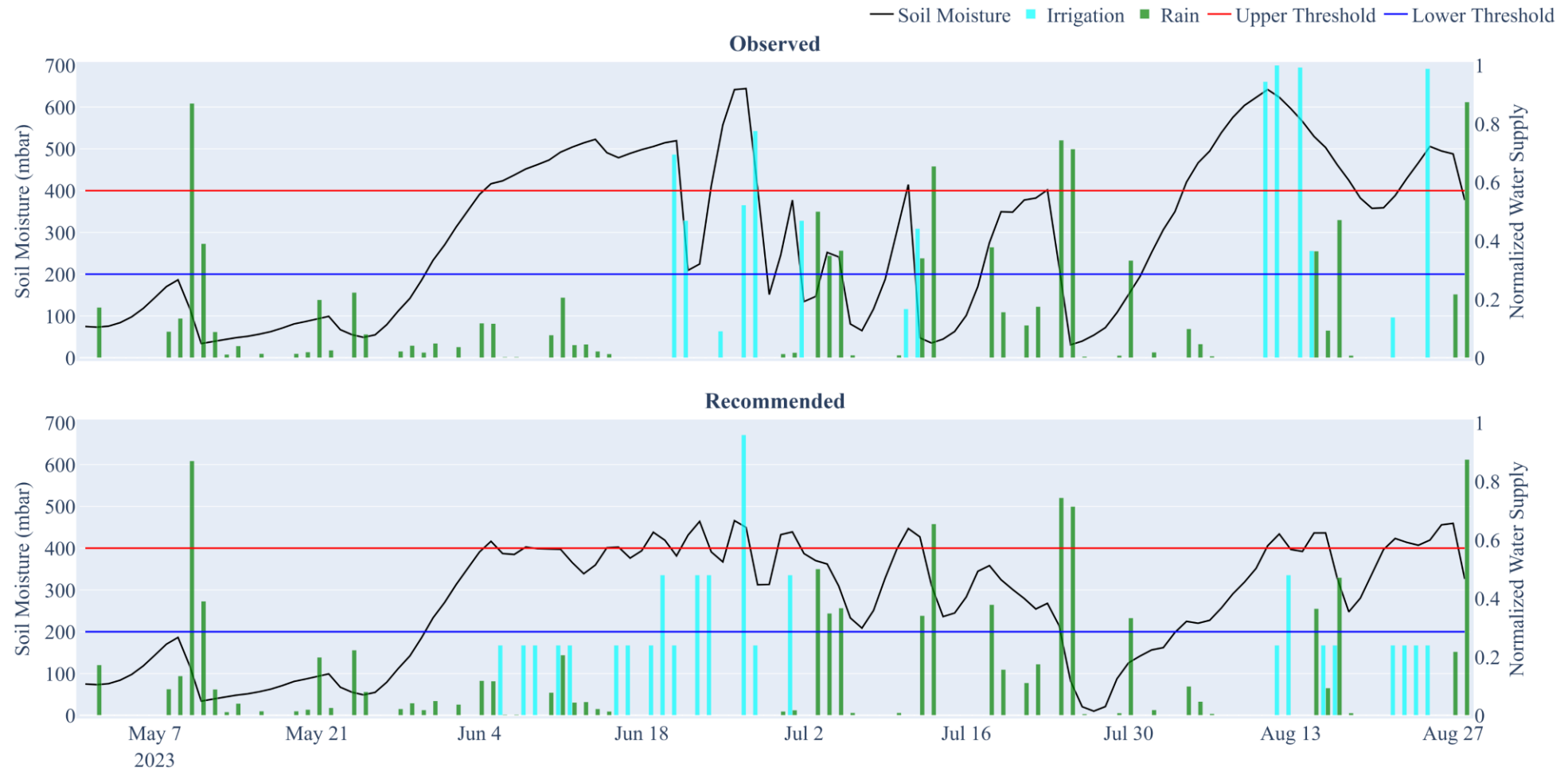


Figure: Comparison of actual and DSS-simulated irrigation schedules for a specific water sector in RDL during 2023

## Results

### Overall Impact on Water Use and Soil Moisture

| Sector       | Water Volume per Row (L) |        |                    | Critical Days |     |                    | Avg Tensiometer (mBars) |       |                    |
|--------------|--------------------------|--------|--------------------|---------------|-----|--------------------|-------------------------|-------|--------------------|
|              | Real                     | DSS    | DSS <sub>opt</sub> | Real          | DSS | DSS <sub>opt</sub> | Real                    | DSS   | DSS <sub>opt</sub> |
| 1            | 1'828                    | 325    | 325                | 3             | 1   | 1                  | 110                     | 196   | 196                |
| 2            | 4'479                    | 4'550  | 1'300              | 9             | 5   | 3                  | 174                     | 184   | 184                |
| 3            | 11'486                   | 5'200  | 5'200              | 29            | 10  | 9                  | 245                     | 254   | 253                |
| 4            | 10'931                   | 10'075 | 9'750              | 44            | 24  | 33                 | 291                     | 266   | 270                |
| 5            | 23'757                   | 7'800  | 8'450              | 29            | 20  | 15                 | 239                     | 272   | 269                |
| <b>Total</b> | 52'501                   | 27'950 | 25'025             | 114           | 60  | 61                 | 1'059                   | 1'172 | 1'172              |

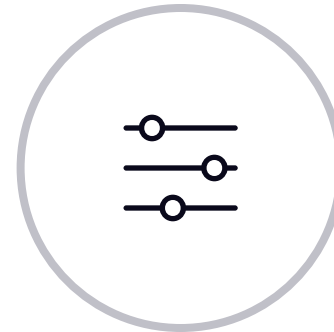
Table: Comparison of irrigation sector performances for RDL during 2023

## Conclusion

### Key Takeaways & Future Directions



Water Savings



Methods



Interpretability



Scalability

*This work was supported by the project “**IRRITRE**: Geographical information system for precision irrigation in Trentino”, funded by the Provincia Autonoma di Trento*

「thank you」